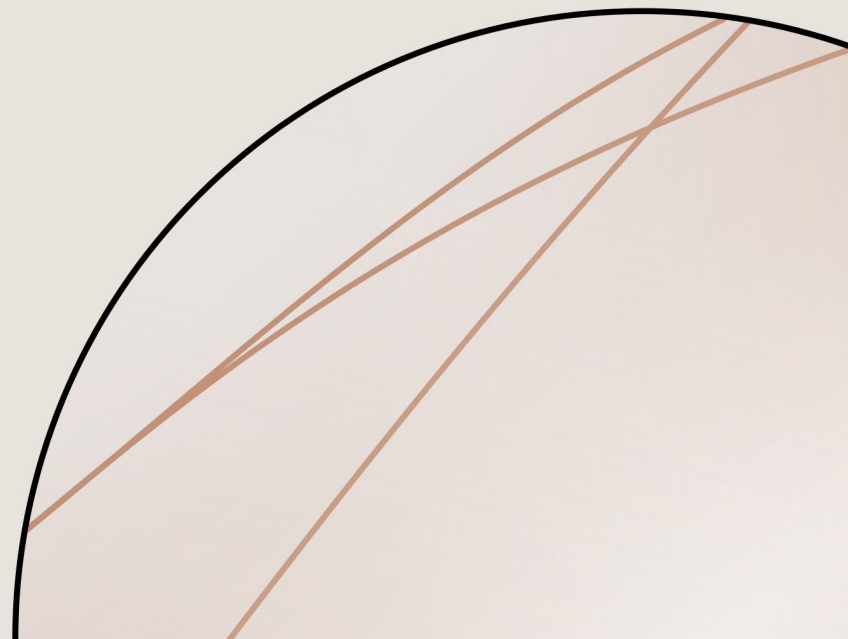


Origins of life in our Solar System

a computational engineering perspective

Benjamin **Terschanski**
PhD defense

October 6th, 2026





How did we get here?
What is the **recipe**?

“Life is self-reproduction
with variations.”

- Trifonov (2012)

“Life is **complex**”

- Terschanski (2026)

life



courtesy of
Dr. Stefan Peters



habitability

may host life



courtesy of
Dr. Stefan Peters

Three ingredients for life (as we know it)



time



liquid water



stability to support the above



courtesy of
Dr. Stefan Peters

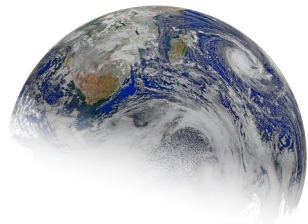


B1-B3



Lammer et al. (2009)

How did we get here?



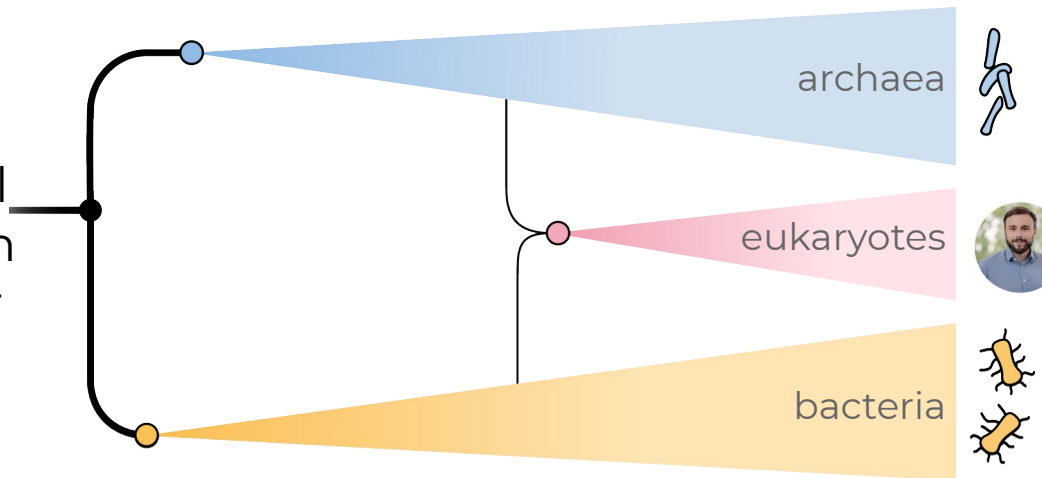
Earth's lifespan (in billion years)



~0.7
life forms

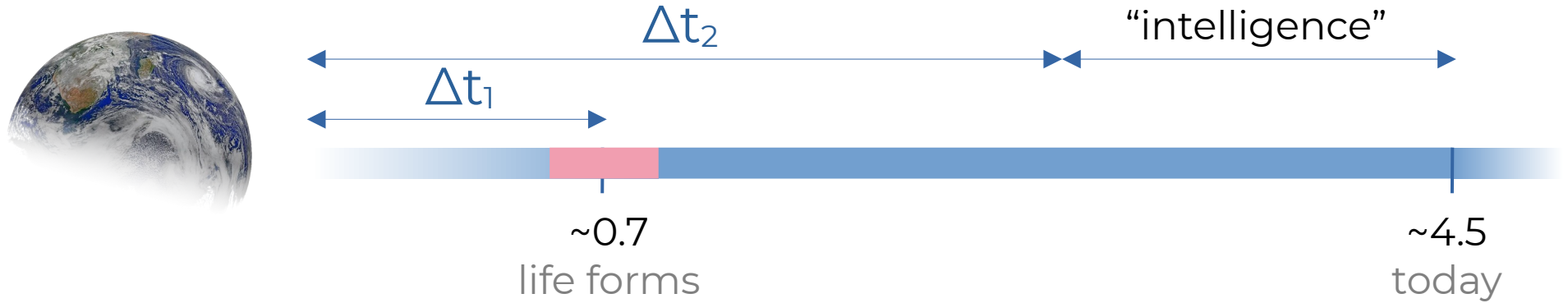
~4.5
today

**Last
Universal
Common
Ancestor**



Are we alone?

Bayesian argument



What is the **life emergence rate λ** per unit time given data "*life emerged early on earth*"?

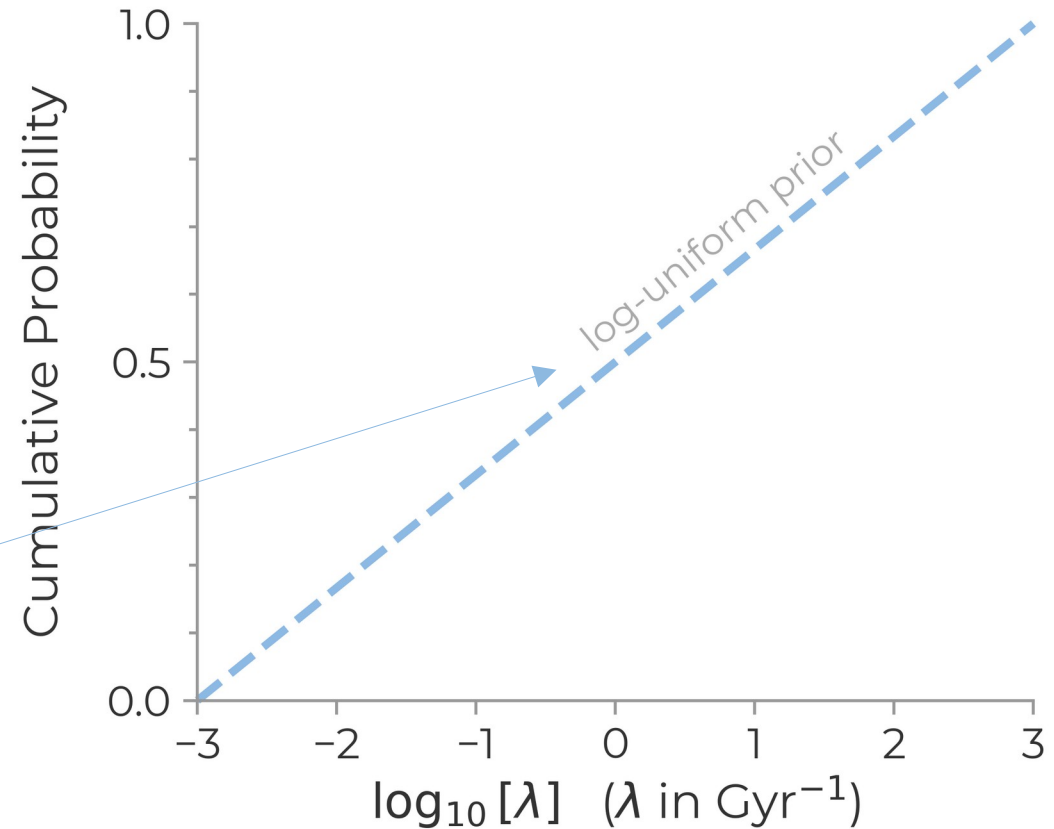
Poisson likelihood
$$P(data|\lambda) = \frac{1 - e^{-\lambda \Delta t_1}}{1 - e^{-\lambda \Delta t_2}}$$

Are we alone?

Estimating life formation rate λ

Posterior

$$P(\lambda|\text{data}) = \frac{\overset{\text{likelihood}}{P(\text{data}|\lambda)} \overset{\text{prior}}{P(\lambda)}}{\underset{\text{normalization}}{P(\text{data})}}$$
$$P(\lambda) = \frac{1}{\lambda(\log(\lambda_{\max}) - \log(\lambda_{\min}))}$$

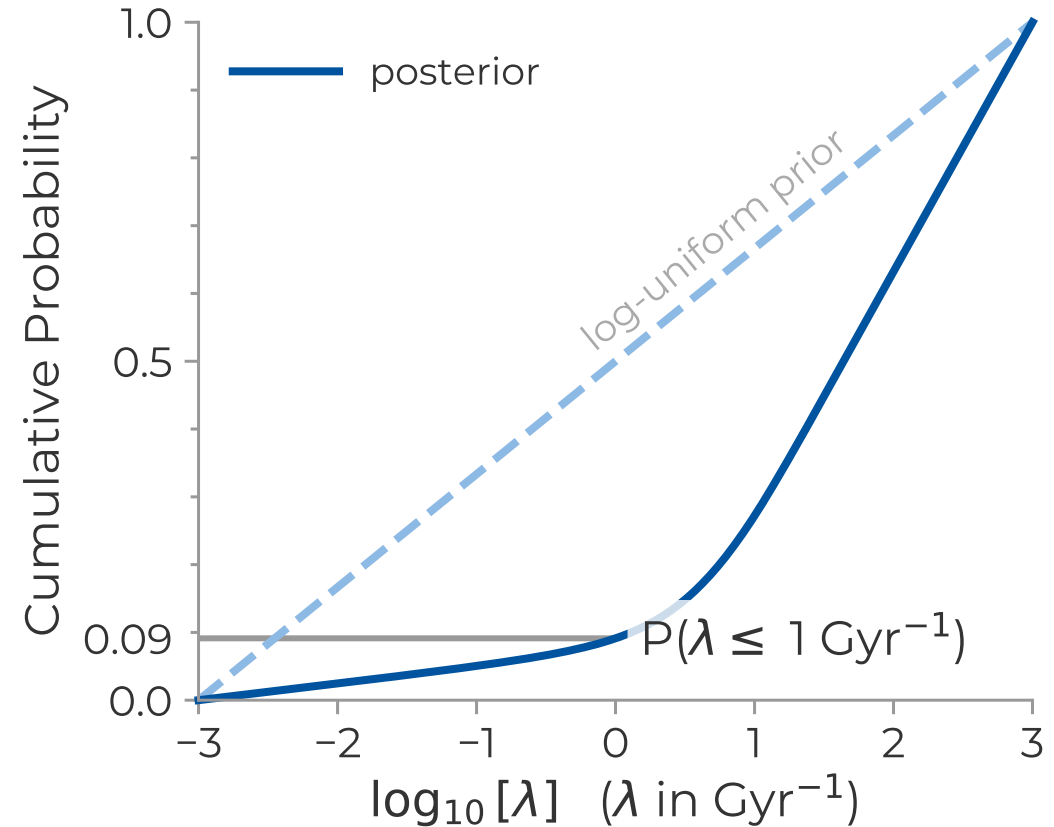


Are we alone?

Estimating life formation rate λ

Posterior

$$P(\lambda|\text{data}) = \frac{\overset{\text{likelihood}}{P(\text{data}|\lambda)} \overset{\text{prior}}{P(\lambda)}}{\underset{\text{normalization}}{P(\text{data})}}$$
$$P(\lambda) = \frac{1}{\lambda(\log(\lambda_{\max}) - \log(\lambda_{\min}))}$$

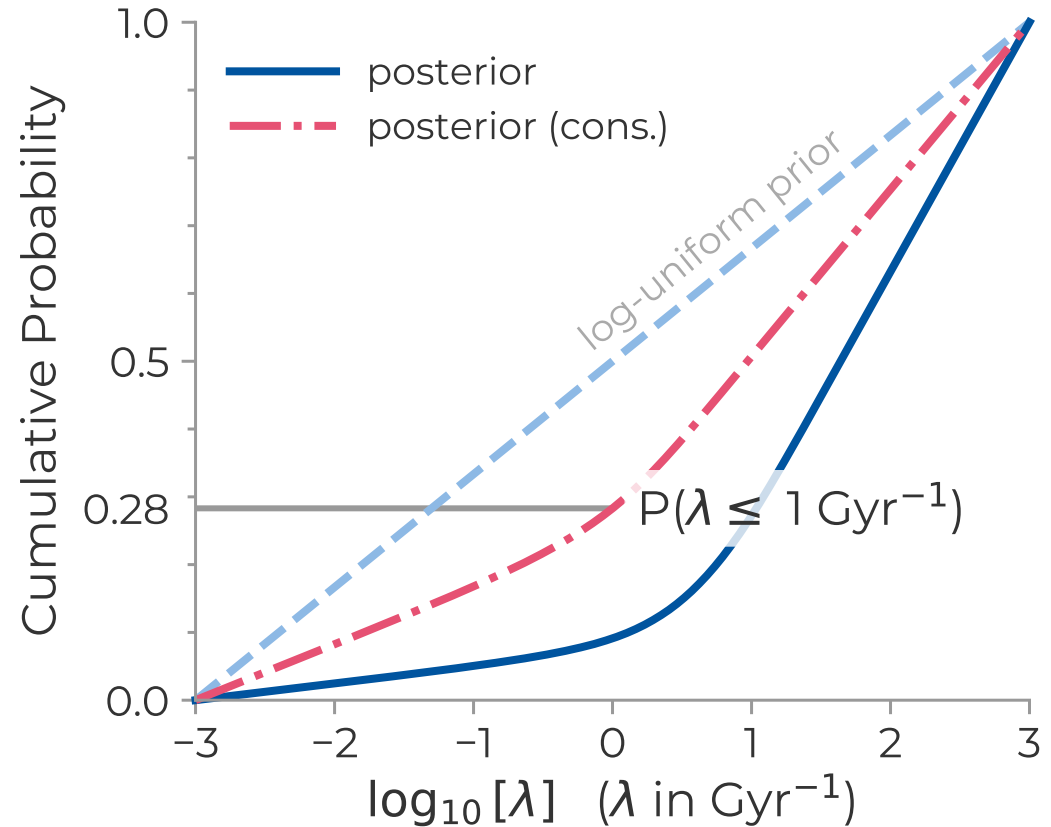
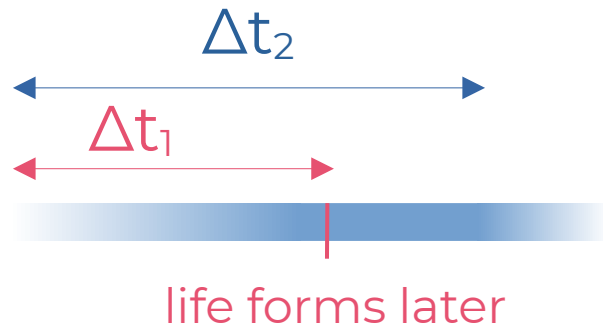


Are we alone?

Estimating life formation rate λ

Posterior

$$P(\lambda|\text{data}) = \frac{\overset{\text{likelihood}}{P(\text{data}|\lambda)} \overset{\text{prior}}{P(\lambda)}}{\underset{\text{normalization}}{P(\text{data})}}$$



Are we alone?

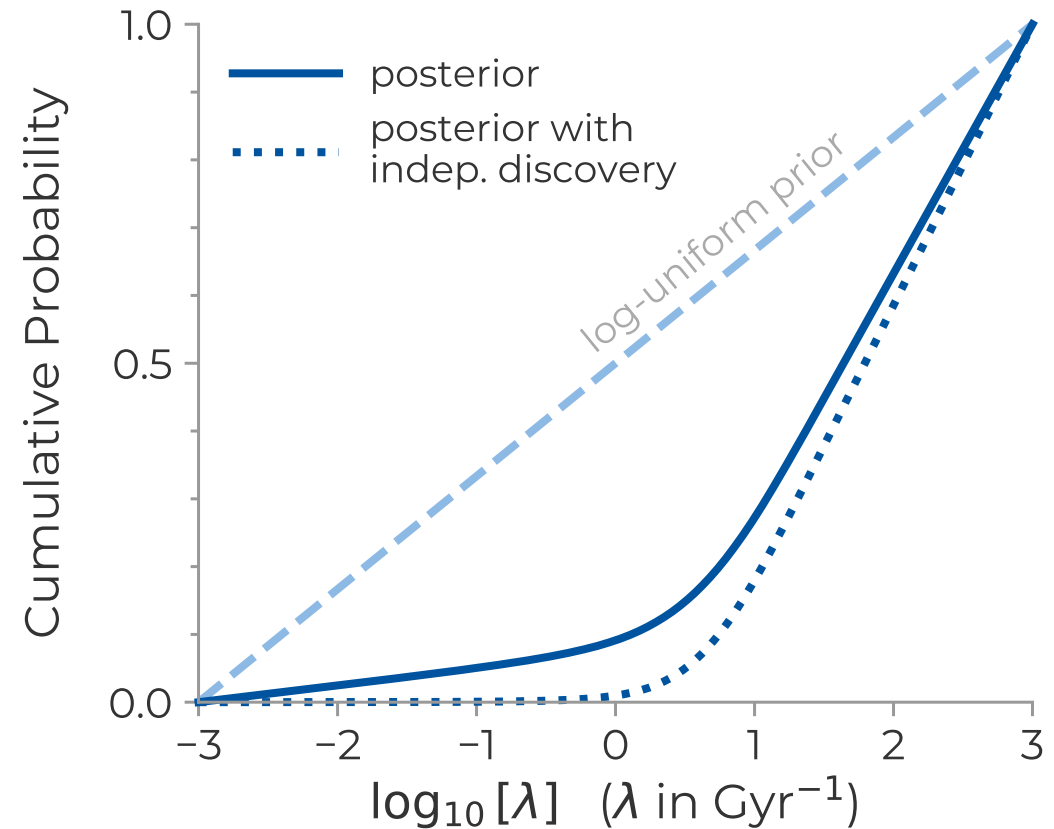
Estimating life formation rate λ

Posterior

$$P(\lambda|\text{data}) = \frac{\overset{\text{likelihood}}{P(\text{data}|\lambda)} \overset{\text{prior}}{P(\lambda)}}{\underset{\text{normalization}}{P(\text{data})}}$$

What if we discovered life on
another planet?

→ **high** λ becomes more likely



Planet Scout24



Quiet blue planet in premium location 

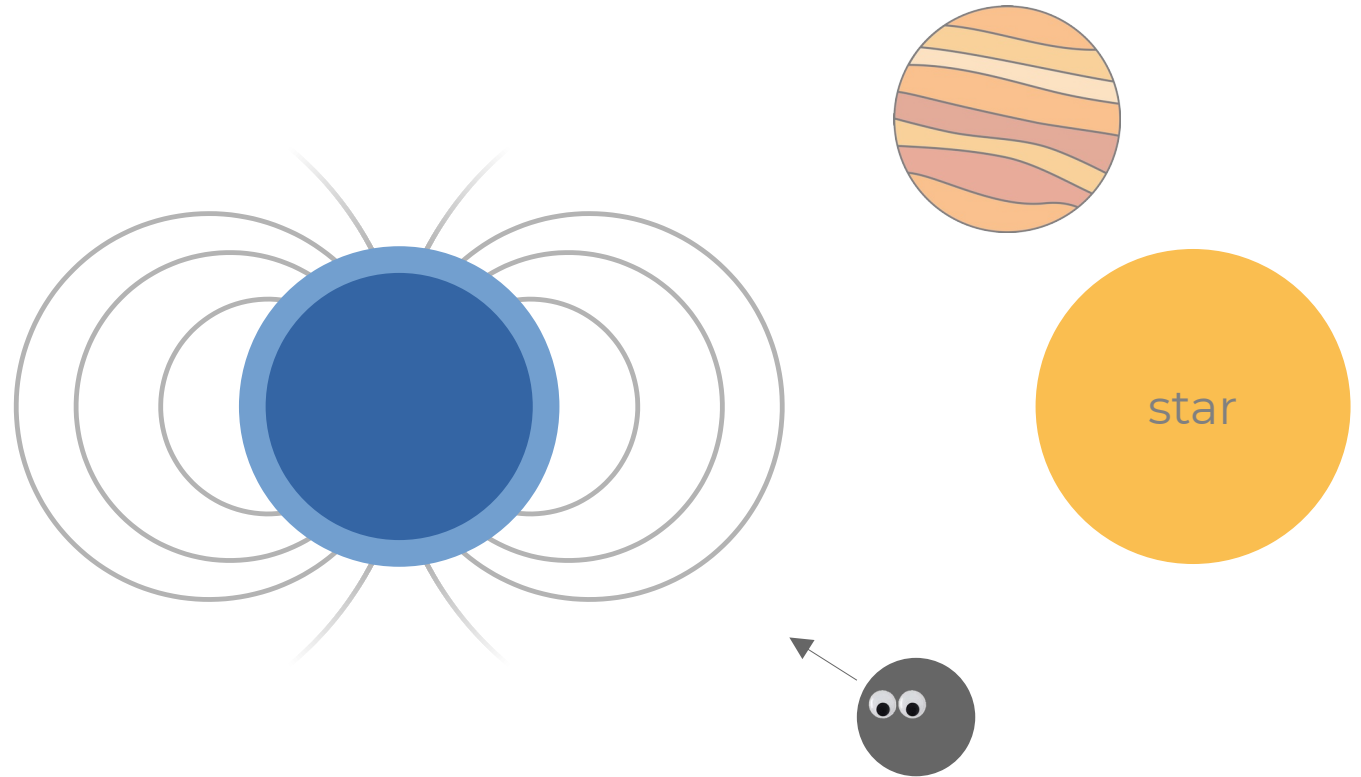
??? € $\sim 10^{12} \text{ km}^3$ single planet

1 AU, Solar System, Milky way

habitability score



1. chemistry
2. energy budget
3. magnetosphere
4. impacts

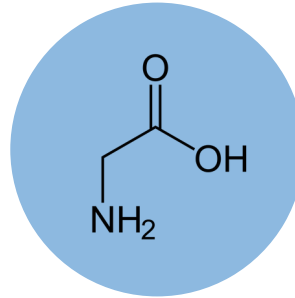


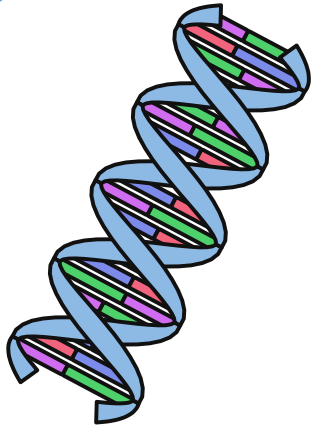
1. chemistry

2. energy budget

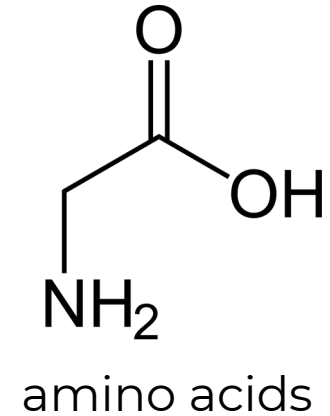
3. magnetosphere

4. impacts



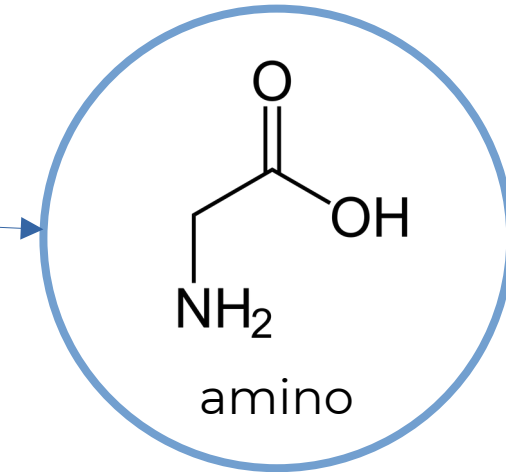
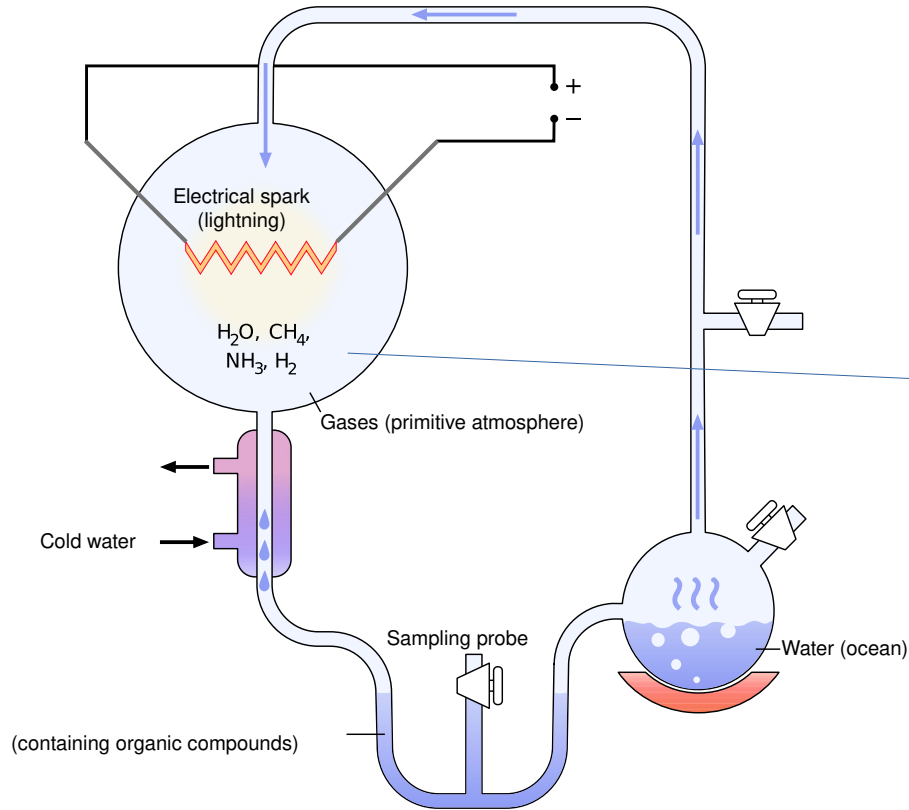


Carbon
Hydrogen
Nitrogen
Oxxygen
Phosphorus
Sulfur



Origin of organic molecules

Miller-Urey experiment



→ **organic compounds** from primordial atmosphere

Redox potential

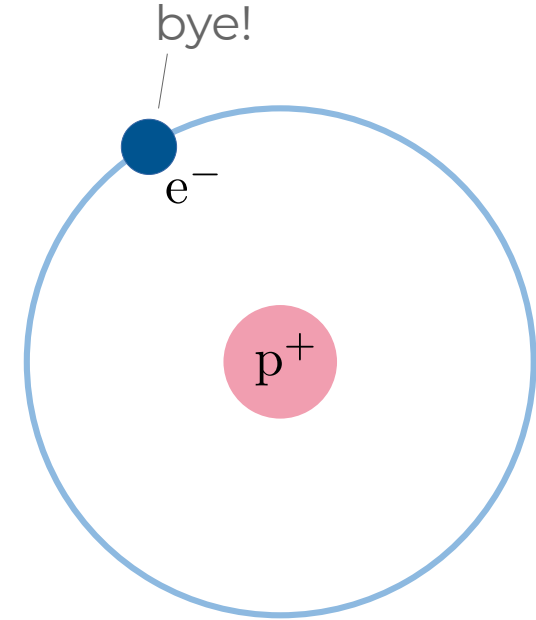
The battery of life

redox reaction $A + B \longrightarrow A^+ + B^-$

reducing agent $A \longrightarrow A^+ + e^-$

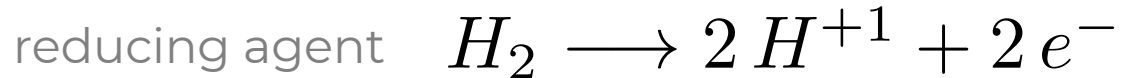
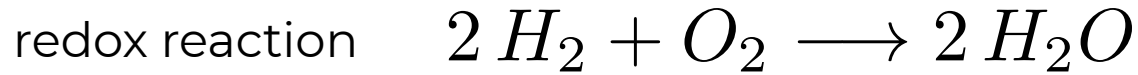
redox potential ⚡

oxidizing agent $B + e^- \longrightarrow B^-$

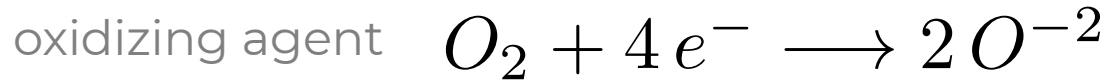


Redox potential

The battery of life

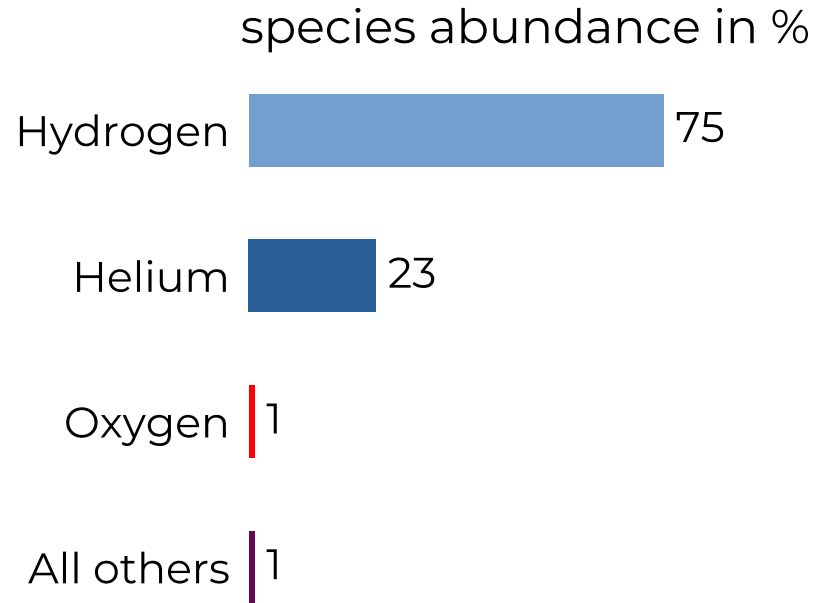
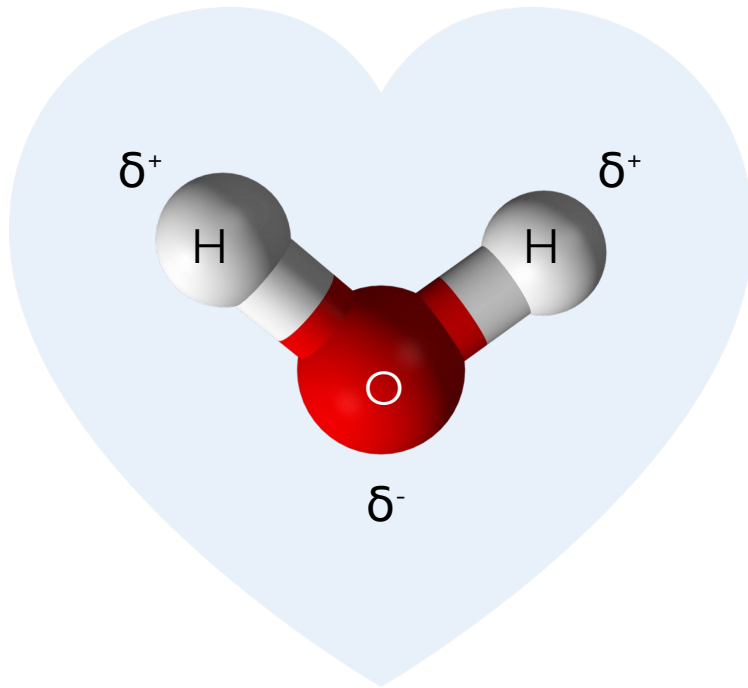


redox potential ⚡

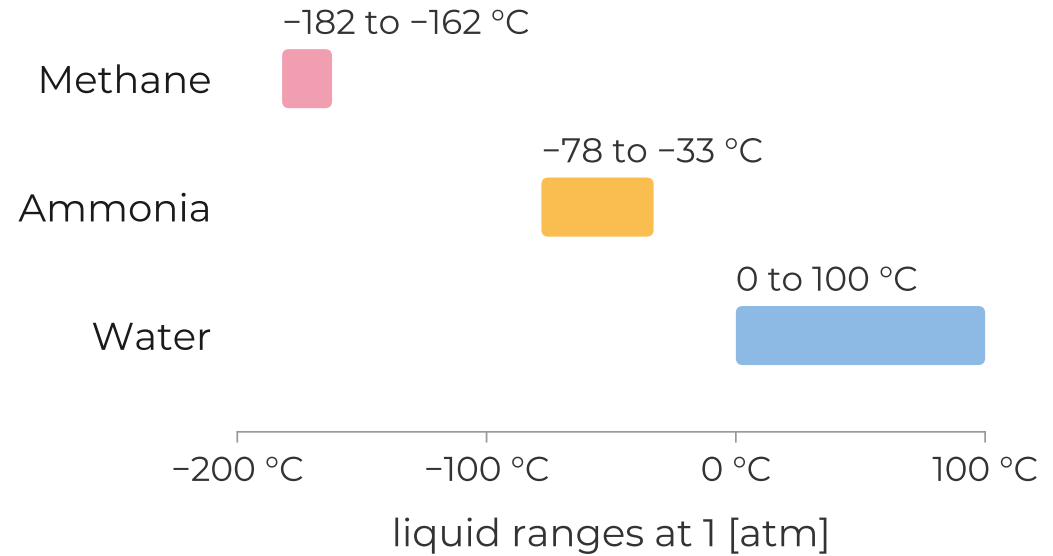
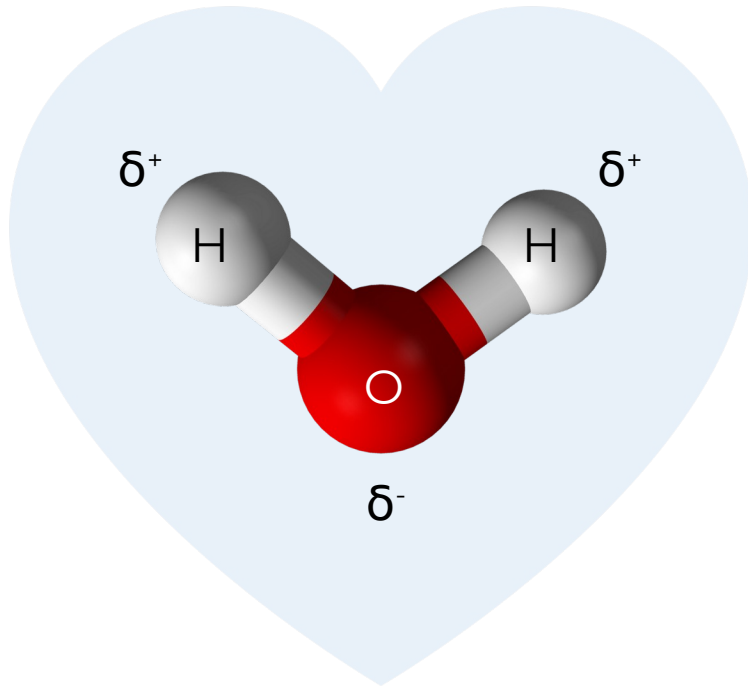


→ works with and without **light** or **oxygen**

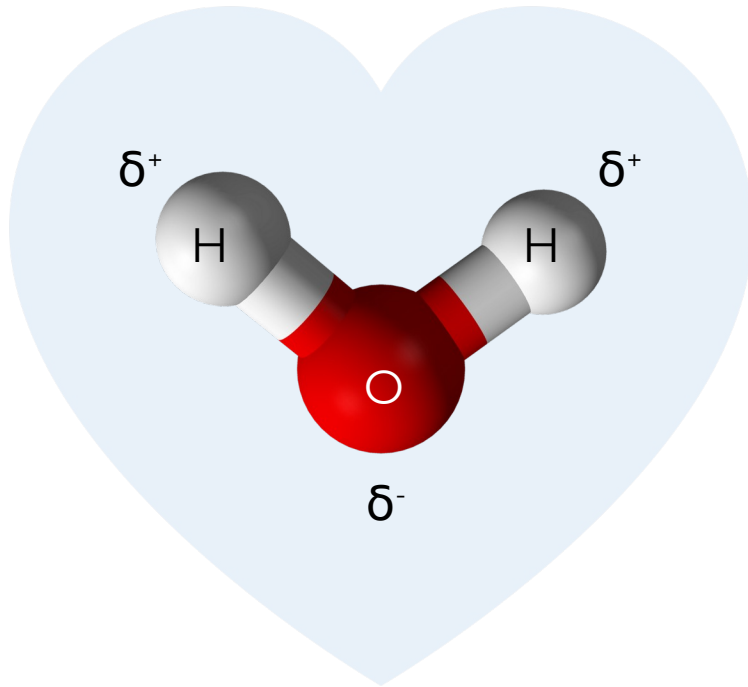
Why liquid water?



Why liquid water?

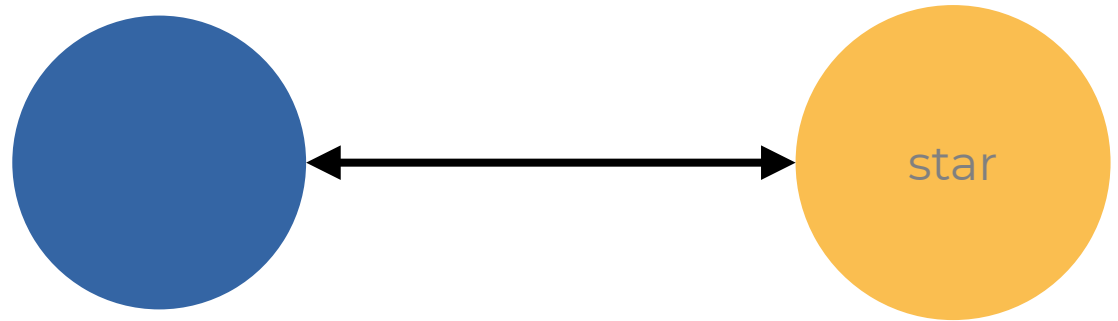


Why liquid water?



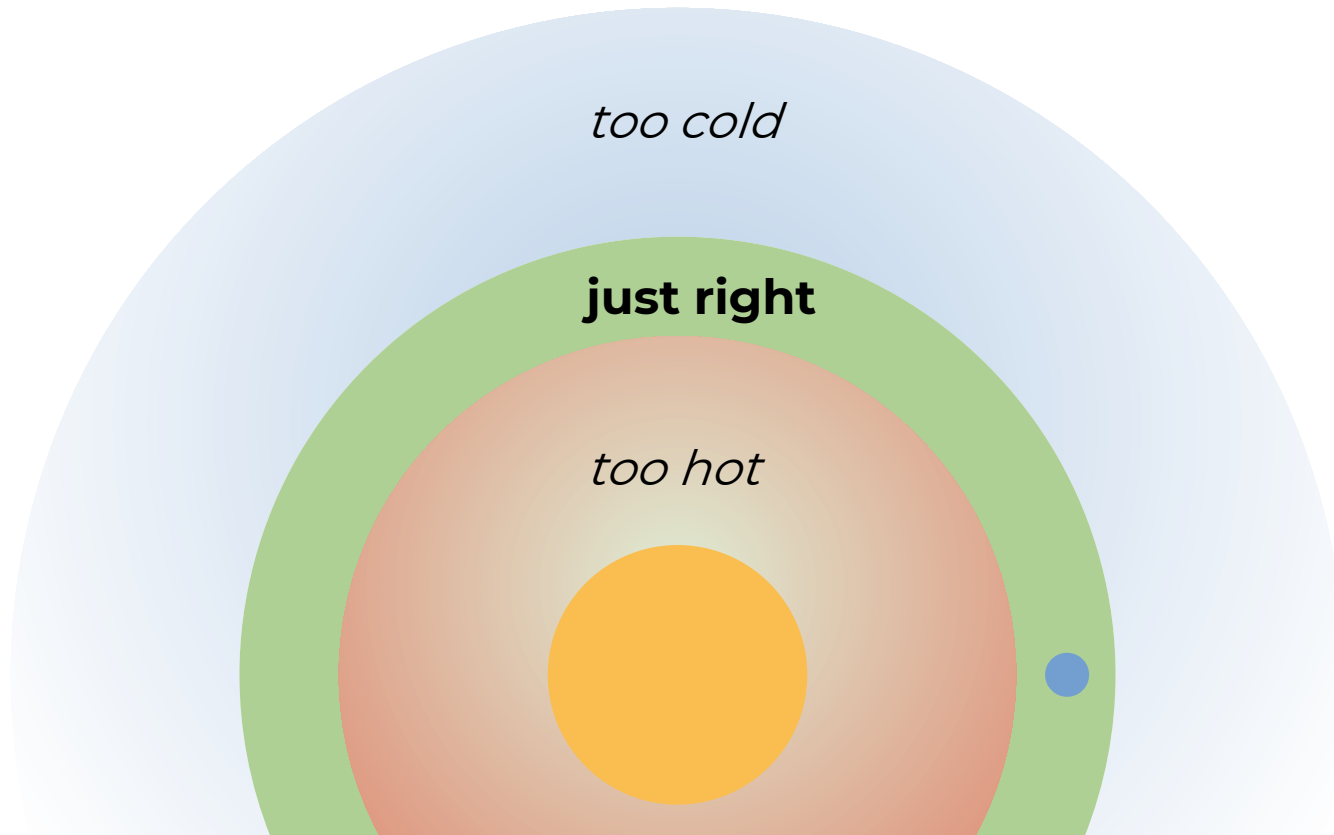
1. good **solvent**
2. good **reaction medium**
3. good temperature **buffer**

1. chemistry
- 2. energy budget**
3. magnetosphere
4. impacts



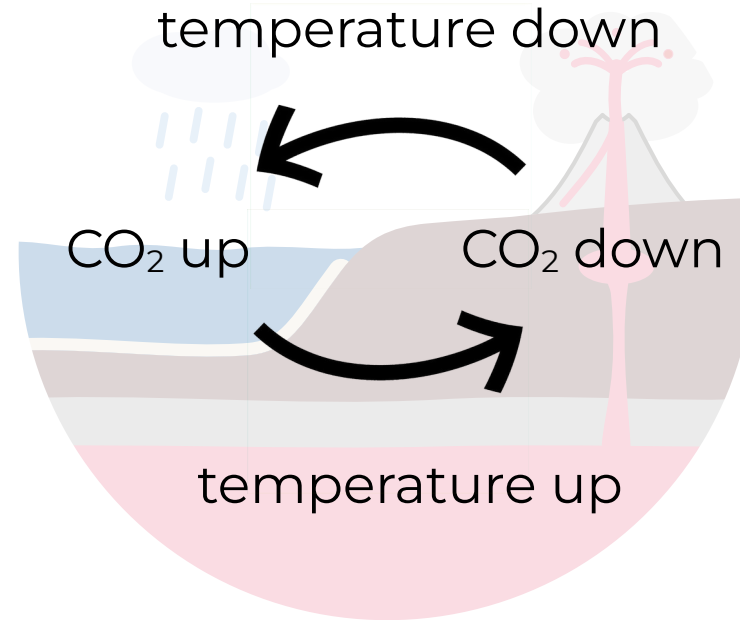
Habitable zone

Conditions for stable surface water

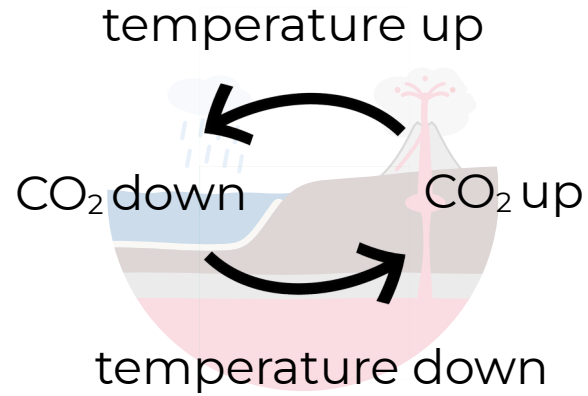


Stable climate feedback

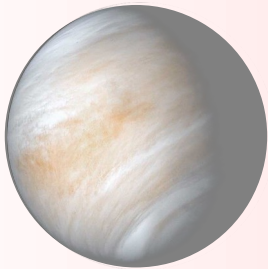
Carbon-Silicate cycle



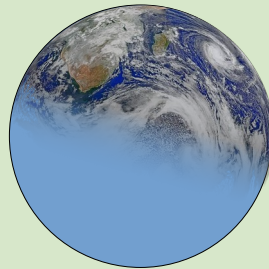
Stable climate feedback



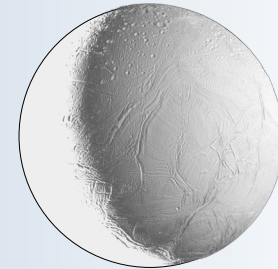
dry
water escapes



habitable planet
stable surface water

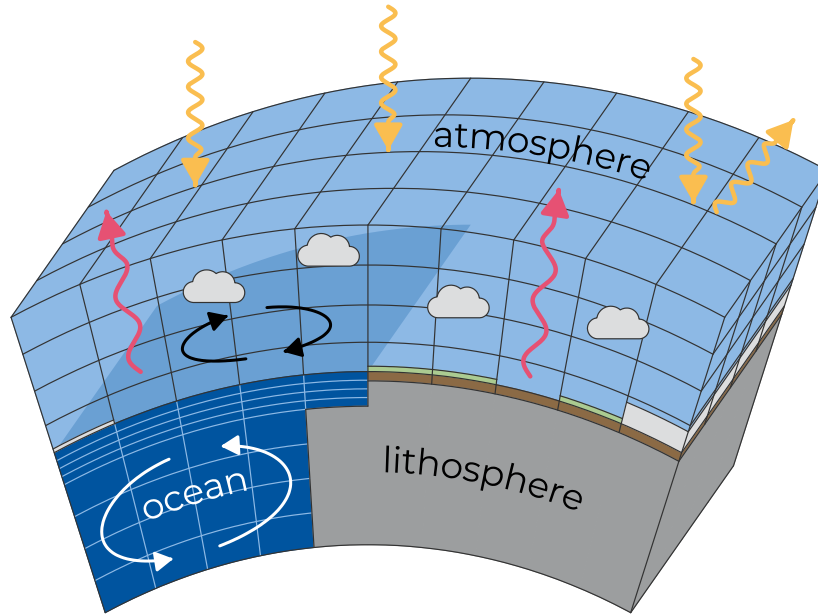


ice
CO₂ condensation



Early Earth's climate

General circulation model



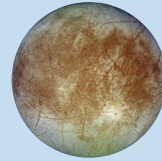
Recreated from: David Bice, Penn State University

3D hydrodynamics
+
radiative transfer
+
condensation model
+
discretization method

Ocean worlds



NASA / Public Domain
Relative scales are approximate

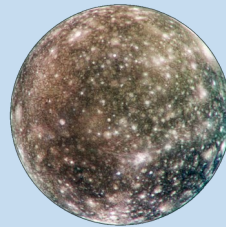


Jovian

Europa



Ganymede



Callisto

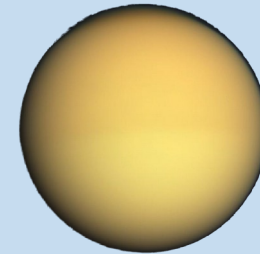
Saturnian



Enceladus



Mimas

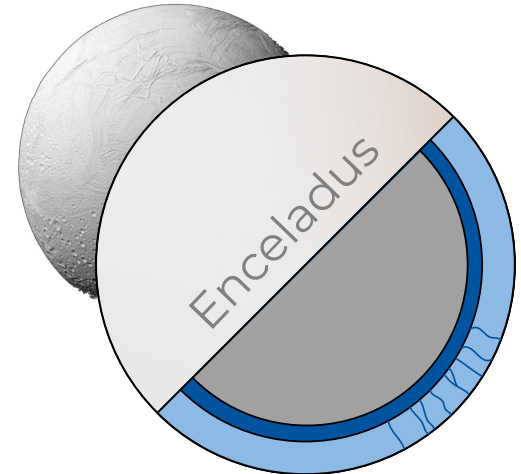
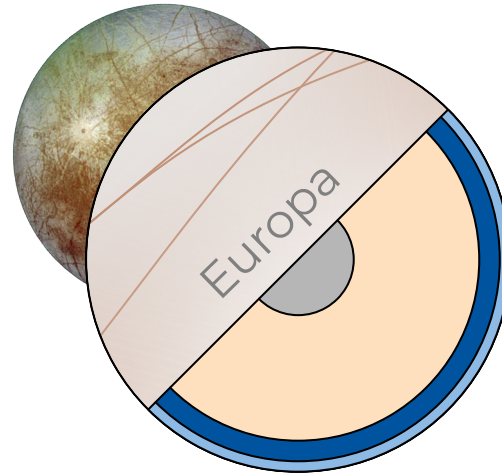
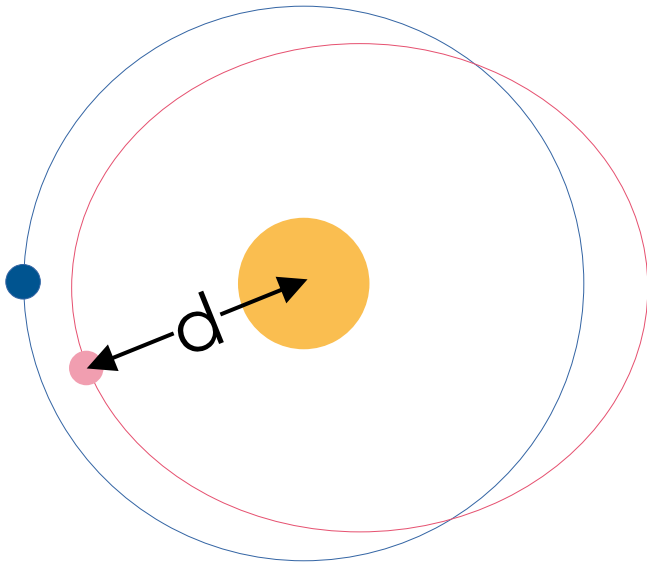


Titan

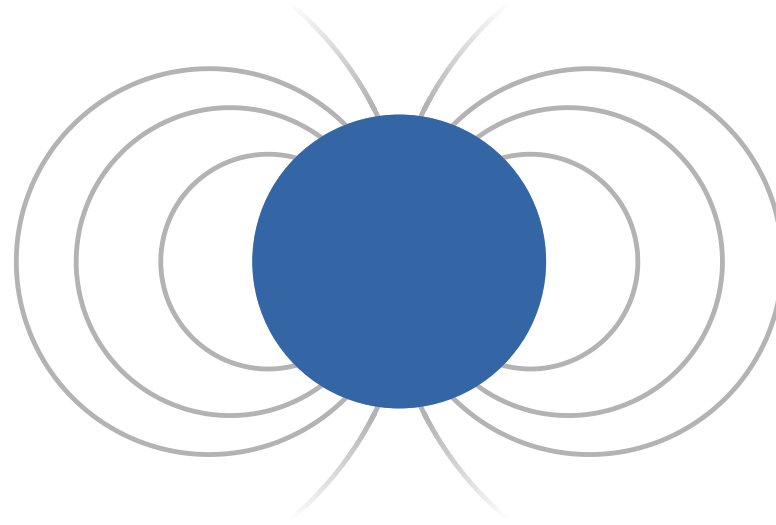
Other heat sources

Tidal heating

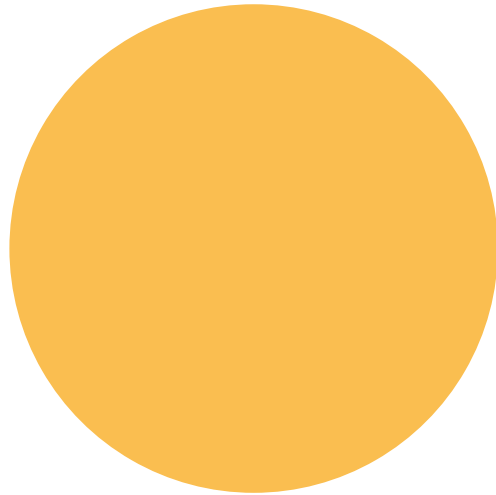
$$\dot{E}_{tidal} = f(\text{orbit, host mass, rheology})$$



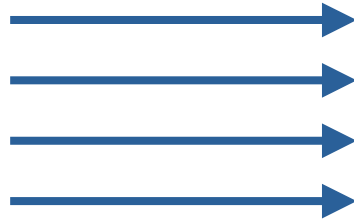
1. chemistry
2. energy budget
- 3. magnetosphere**
4. impacts



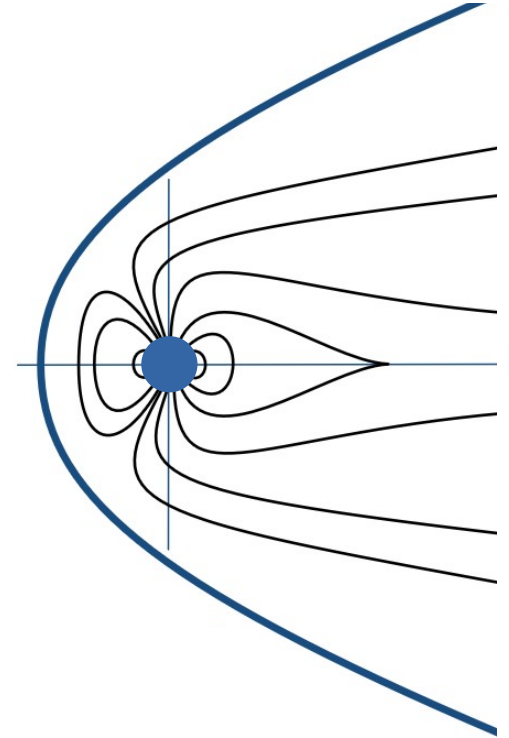
Magnetic field

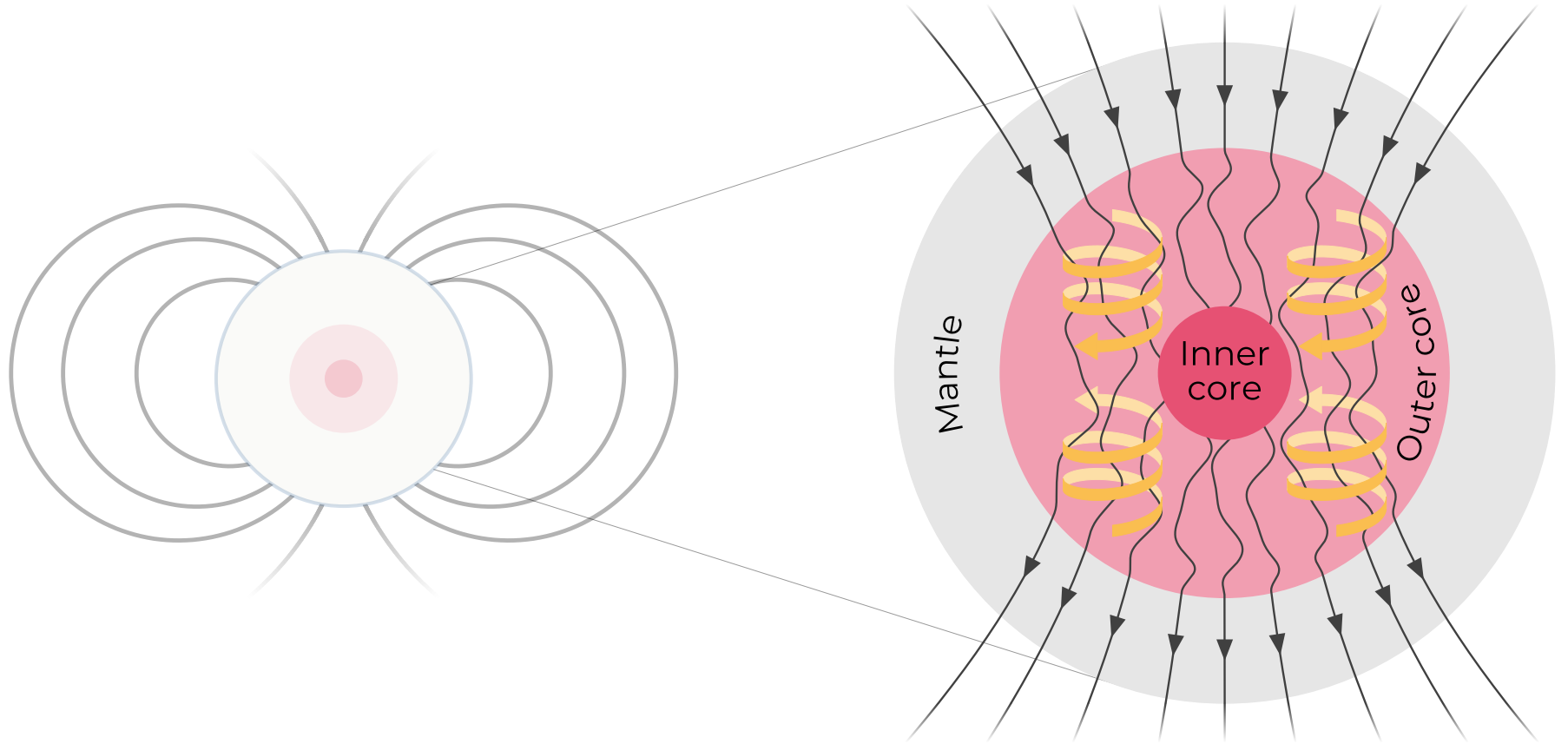


solar wind



strips atmosphere
kills DNA





Induction equation

$$\text{Re}_m \left[\frac{\partial \mathbf{B}}{\partial t} - \nabla \times (\mathbf{v} \times \mathbf{B}) \right] = \nabla^2 \mathbf{B}$$

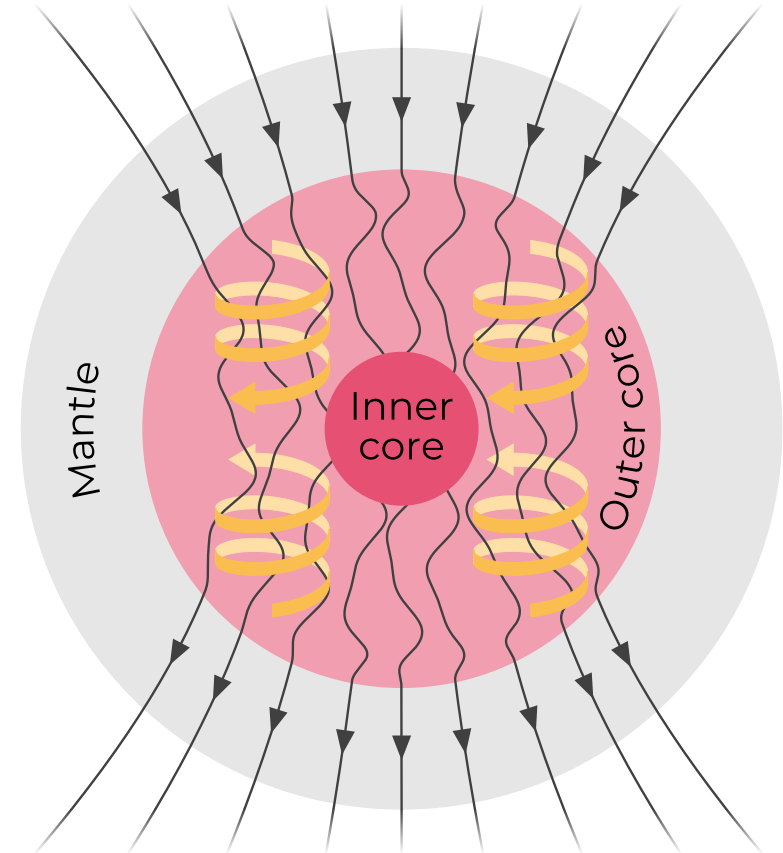
$$\nabla \cdot \mathbf{B} = 0$$

magnetic Reynolds number $\text{Re}_m = \frac{V L}{\nu}$.

electric field $\mathbf{B}$

velocity $\mathbf{v}$

magnetic diffusivity ν



Induction equation

$$\text{Re}_m \left[\frac{\partial \mathbf{B}}{\partial t} - \nabla \times (\mathbf{v} \times \mathbf{B}) \right] = \nabla^2 \mathbf{B}$$

$$\nabla \cdot \mathbf{B} = 0$$

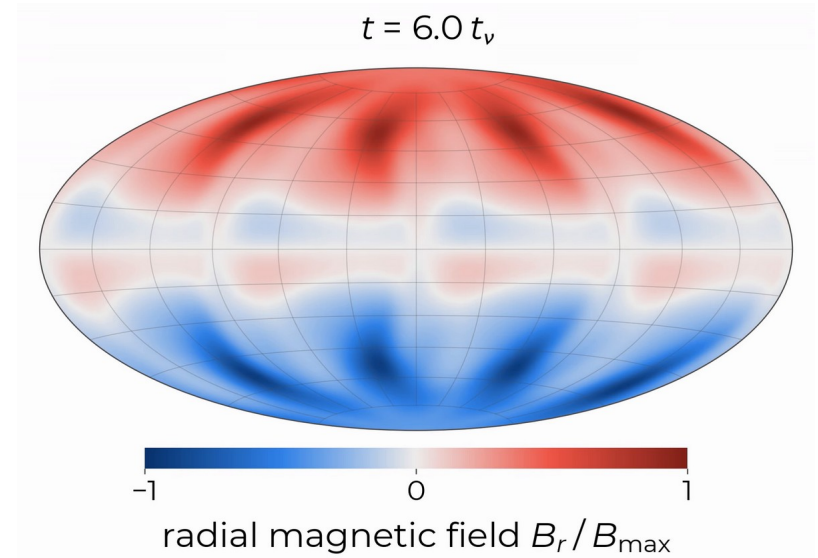
magnetic Reynolds number $\text{Re}_m = \frac{V L}{\nu}$.

electric field $\mathbf{B}$

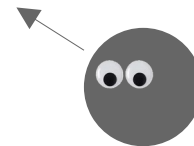
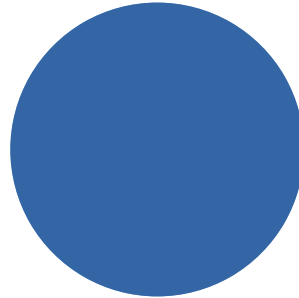
velocity $\mathbf{v}$

magnetic diffusivity ν

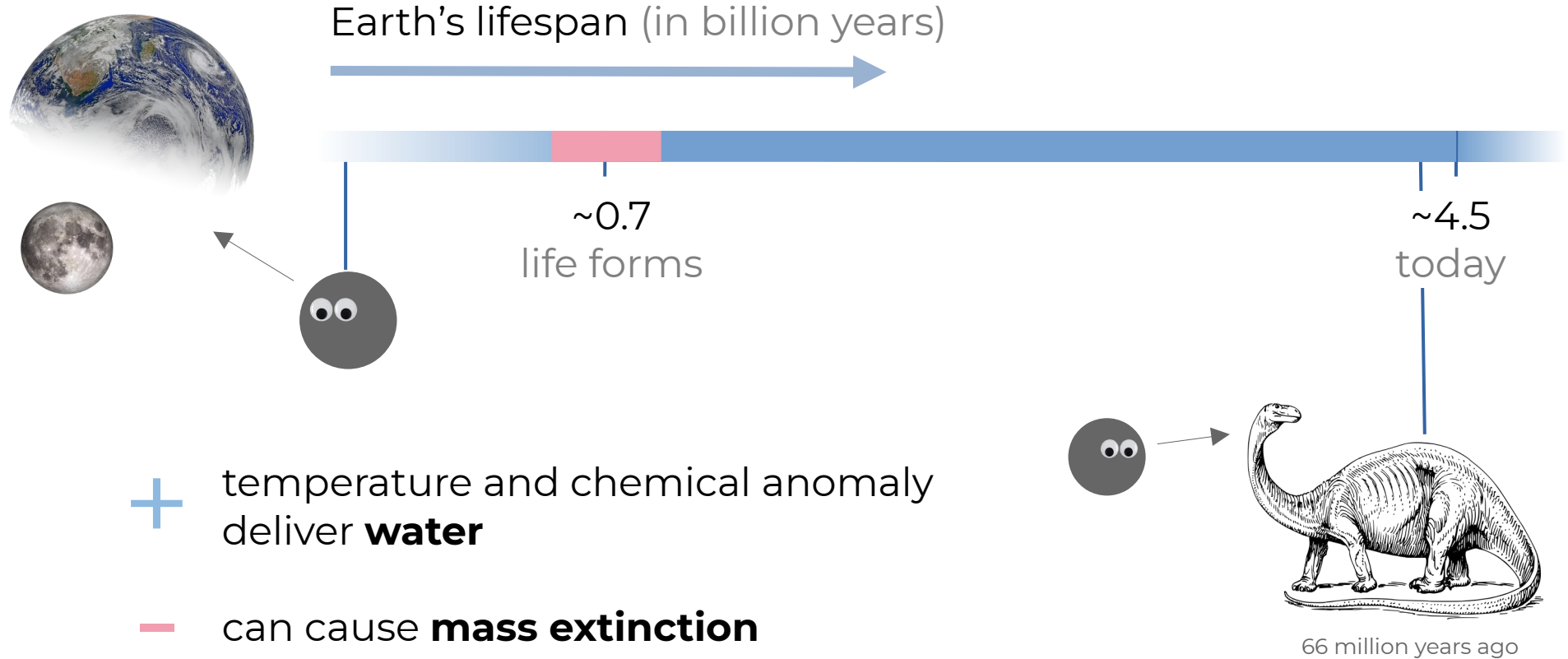
nschaeff.bitbucket.io/xshells/

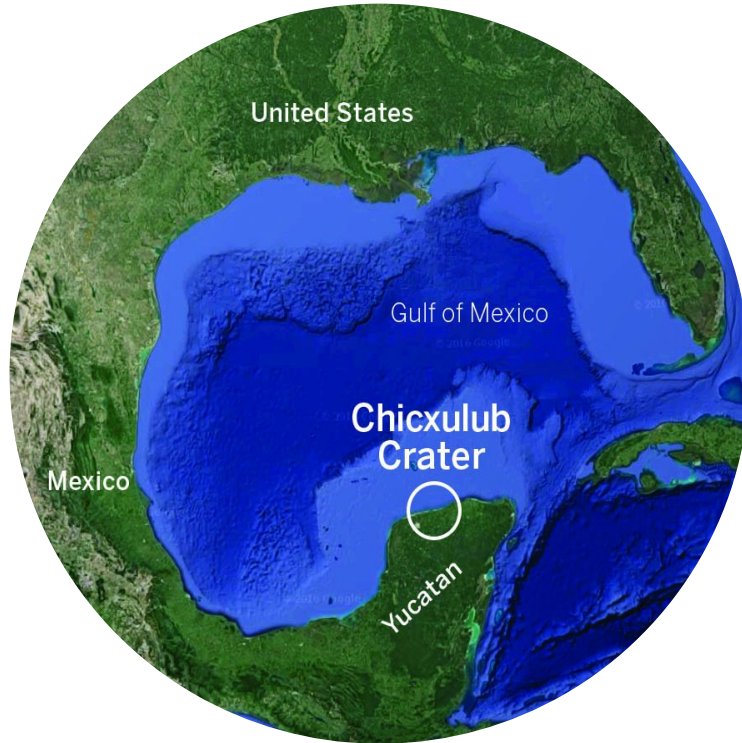


1. chemistry
2. energy budget
3. magnetosphere
- 4. impacts**

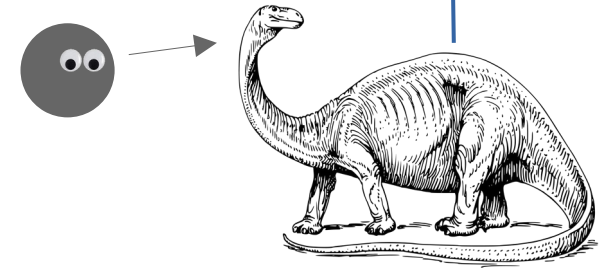


Impacts



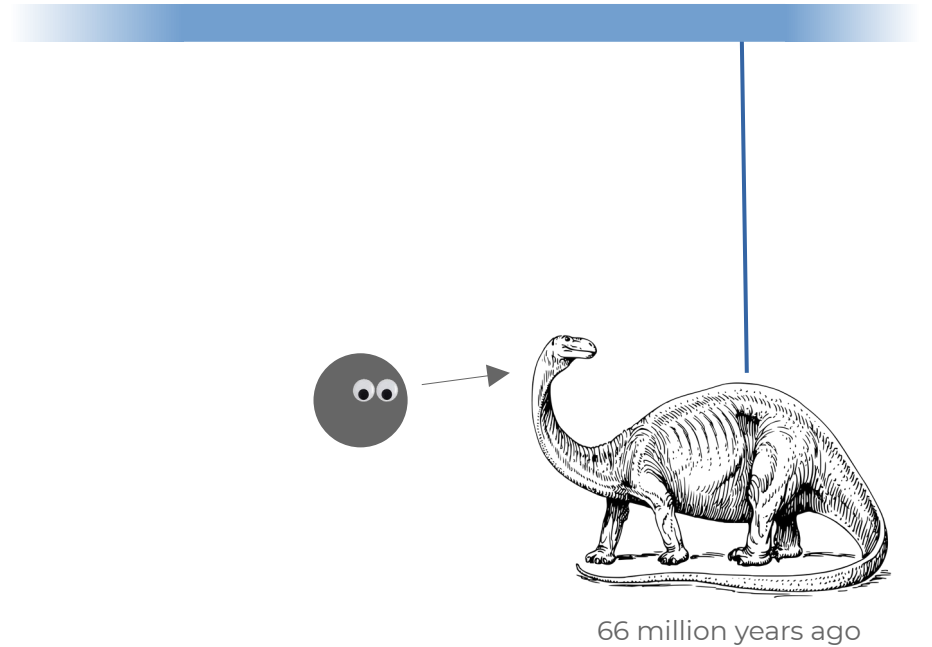
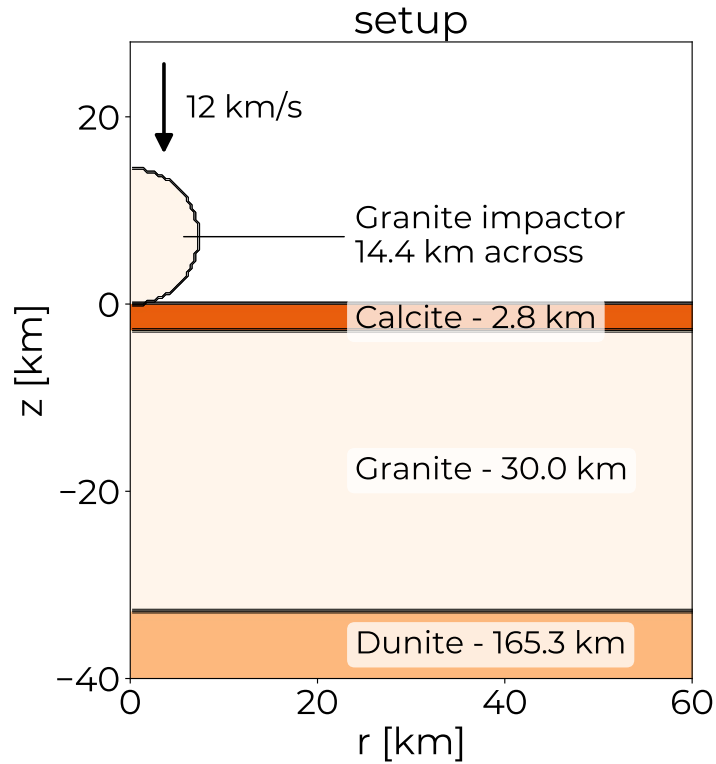


Google Maps / UT Jackson School of Geosciences



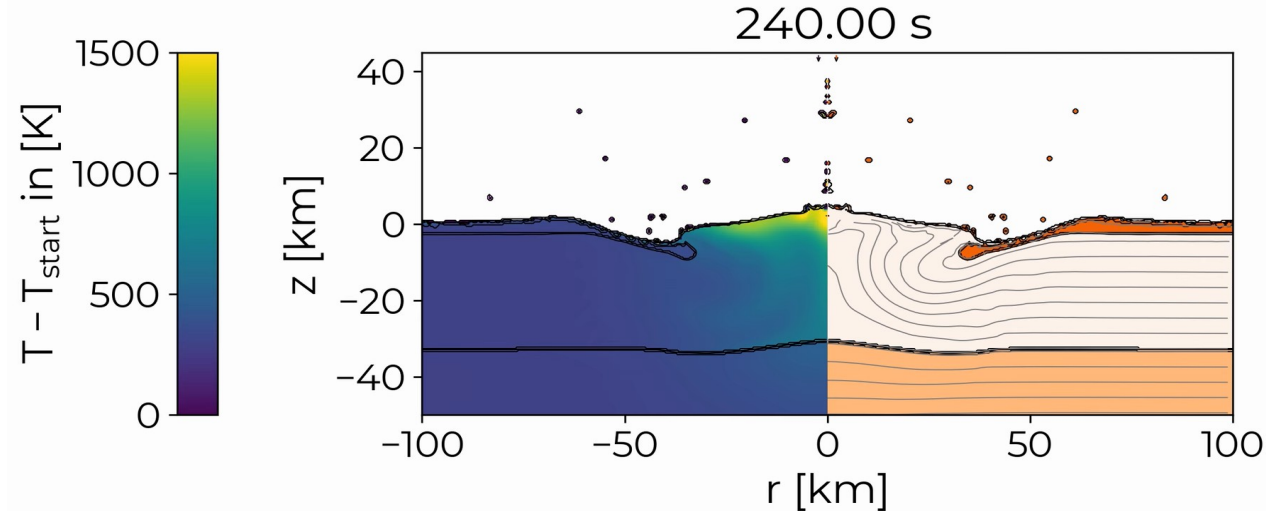
66 million years ago

Impacts



The Chicxulub impact

Numerical simulations



github.com/isale-code/isale-wiki/wiki/Chicxulub-crater-formation

~3h runtime on my i7-10510U CPU @ 1.80GHz



Planet Scout24

1. chemistry
2. energy budget
3. magnetosphere
4. impacts



Chill moon with a Jupiter view

??? € $\sim 1.6 \times 10^{10} \text{ km}^3$ icy moon
5.2 AU, Solar System, Milky way



habitability score





Planet Scout24



Chill moon with a Jupiter view

??? € ~1.6x10¹⁰ km³ icy moon

5.2 AU, Solar System, Milky way



habitability score



Planet Scout24



Quiet blue planet in premium location



??? € $\sim 10^{12} \text{ km}^3$ single planet

1 AU, Solar System, Milky way



Thank you for your attention.

Slides, figures and references at
github.com/benjaminte/phd_defense



Several anonymous reviewers helped to improve this
talk ❤️

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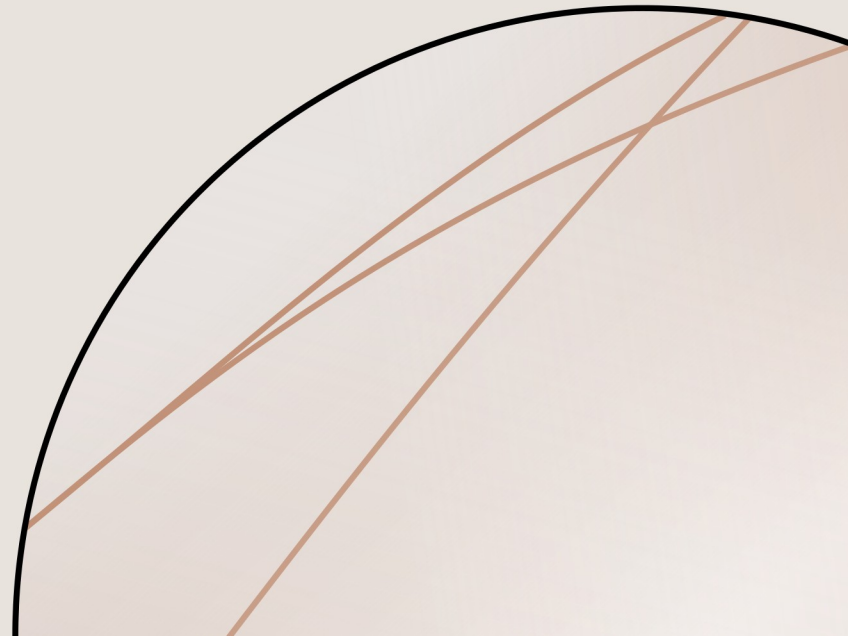
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Images

- B1) "Earth from space." Ocean Biology Processing Group, NASA Goddard Space Flight Center.
- B2) "Hadean Earth." Simone Marchi (SwRI), SSERVI, NASA. Astronomy Picture of the Day, August 5, 2014. <https://apod.nasa.gov/apod/ap140805.html>.
- B3) "Butterfly." Stefan Peters. Private collection.
- B4) "Habitability Venn diagram." C. S. Cockell et al. "Habitability: A Review," *Astrobiology* 16, no. 1 (2016): 89–117, Fig. 3, after T. M. Hoehler, *Astrobiology* 7, no. 6 (2007): 824–38. © 2016 Mary Ann Liebert, Inc. Redrawn with modifications. <https://doi.org/10.1089/ast.2015.1295>.
- B5) "Ball-and-stick model of the water molecule." Ben Mills. Public domain. <https://commons.wikimedia.org/wiki/File:Water-3D-balls-A.png>.
- B6) "General circulation model, concept sketch." David Bice, Penn State University. EARTH 103: Earth in the Future. CC BY-NC-SA 4.0. Redrawn with modifications. <https://courses.ems.psu.edu/earth103/node/524>.
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- B8) "Northern lights over Pinehouse Lake." Dreerwin. CC BY-SA 4.0. https://commons.wikimedia.org/wiki/File:Northern_Lights,_Pinehouse.jpg.
- B9) "Geodynamo sketch." Andrew Z. Colvin, after the US Geological Survey. CC BY-SA 4.0. Redrawn with modifications. https://commons.wikimedia.org/wiki/File:Dynamo_Theory_-_Outer_core_convection_and_magnetic_field_generation.svg.
- B10) "Miller–Urey experiment." YassineMrabet. CC BY-SA 3.0. https://commons.wikimedia.org/wiki/File:Miller-Urey_experiment-en.svg.



Images (continued)

- B11) "Ball-and-stick model of adenosine triphosphate (ATP)." Ben Mills. Public domain. <https://commons.wikimedia.org/wiki/File:ATP-3D-balls.png>.
- B12) "Tree of life showing the last universal common ancestor (LUCA)." Encyclopædia Britannica, Inc. <https://www.britannica.com/science/last-universal-common-ancestor>.
- B13) "Moon." NASA. Public domain.
- B14) "Dinosaur sketch." Pearson Scott Foresman. Public domain. [https://commons.wikimedia.org/wiki/File:Brontosaurus_\(PSF\).jpg](https://commons.wikimedia.org/wiki/File:Brontosaurus_(PSF).jpg).
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